Khang B. Tran

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SUMMARY

Third-year B.S. Computer Science student and QA Intern targeting **QA Engineer / SDET** roles. Wrote and ran a 142-case regression suite (426 executions across 3 environments) for a fintech trading web app. Built 1,200+ automated tests across 5 projects using Pytest, JUnit, Mockito, Selenium, and Testcontainers, with GitHub Actions CI.

EXPERIENCE

Quality Assurance Intern | Investagrams Inc. | Remote | Sep 2026 – Present

- Wrote 142 step-by-step regression cases with expected results, covering 13 InvestaTrade web-app modules (KYC, portfolio, Explore, watchlist, cash-in).
- Ran the full suite across 3 environments (426 executions), reaching a 100% final pass rate before release.
- Prioritized testing by risk: 71 High / 59 Medium / 12 Low cases, so the highest-impact flows were covered first.
- Authored 50 of the 142 cases (21 Explore, 17 stock-detail, 12 KYC validation) using boundary-value and negative-input testing.

PROJECTS

FlowOps: IT Workflow Automation Platform | Python, FastAPI, PostgreSQL | github.com/KhangTran69/flowops

- Built a multi-tenant IT ticket and onboarding/offboarding automation platform with 100+ API routes, 13 Alembic migrations, and a Jinja2 operator console.
- Prevented duplicate processing on retried requests using fenced idempotency keys, leased worker jobs with backoff, and dead-letter requeue; verified by 478 Pytest tests plus PostgreSQL reliability tests in GitHub Actions CI.
- Added SLA-risk and time-saved reporting through tenant-scoped analytics and SQL KPI views, with rule-based routing and human-approved AI triage; Microsoft Graph and ConnectWise clients were tested against simulators.

IT Help Desk & Asset Management | Java, Spring Boot, MySQL | github.com/KhangTran69/it-help-desk

- Delivered employee and technician workflows for tickets and assets, with 2 enforced roles and a 4-stage ticket lifecycle.
- Secured ticket data with BCrypt, CSRF protection, JWT bearer auth, and owner-scoped access; optimistic locking rejects stale edits.
- Verified core flows with 110+ automated tests in GitHub Actions CI: JUnit/Mockito unit tests, Testcontainers MySQL/Redis integration tests, and a Selenium browser test.

PH SMS Spam & Scam Detector | Python, scikit-learn, FastAPI, Docker | github.com/KhangTran69/sms-spam-detector

- Built an English/Filipino/Taglish SMS spam classifier (TF-IDF plus URL, phone, money, and urgency features in Logistic Regression) trained on public UCI data plus synthetic PH messages, reaching 94.6% precision, 93.9% recall, and 0.943 F1 on an 810-message held-out test split (5,670 total), versus 0.855 F1 for a Naive Bayes baseline.
- Served predictions through a Dockerized FastAPI API and web app with per-message risk signals; verified by 460 Pytest tests.

Garbage Route Optimizer (Thesis) | Python | Mar–May 2026 | github.com/KhangTran69/angeles-city-garbage-optimizer

- Built a capacity-aware garbage-truck route planner for up to 33 Angeles City barangays using a Genetic Algorithm with 2-opt local search on OpenStreetMap road distances (Haversine fallback).
- Made results reproducible with a fixed-seed benchmark against a greedy baseline and 74 Pytest tests; added experiment history, Folium truck maps, and CSV/ZIP exports with fuel, cost, and CO2 estimates.

PoPify: E-commerce Operations Automation | Python, FastAPI, PostgreSQL, n8n | github.com/KhangTran69/popify

- Automated 3 workflows (order intake, support triage, delivery-delay detection) with human-approved mock actions and blocked blind retries on ambiguous outcomes; verified by 113 automated tests and a six-worker PostgreSQL concurrency check in CI.

SKILLS

QA & Testing: Manual, regression, functional, and smoke testing; test case design; risk-based, boundary-value, and negative testing; bug reports
Test Automation: Pytest, JUnit 5, Mockito, Selenium WebDriver, Testcontainers, API testing (REST, Swagger/OpenAPI)
Languages: Python, Java, SQL, JavaScript, HTML/CSS
Databases & Tools: PostgreSQL, MySQL, SQLite, Redis, Git/GitHub, GitHub Actions (CI/CD), Docker, Docker Compose

EDUCATION

City College of Angeles | B.S. Computer Science | Angeles City, Pampanga | 2024 – 2028 (expected)
GPA: 1.75 (Philippine scale, 1.00 = highest) | Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management

ACTIVITIES

Republic Central Colleges Documentation Team, Business Manager / Videographer (2021 – 2023): produced 15 school videos (41,575 views) in Adobe Premiere Pro and grew the Facebook following by 2,000.